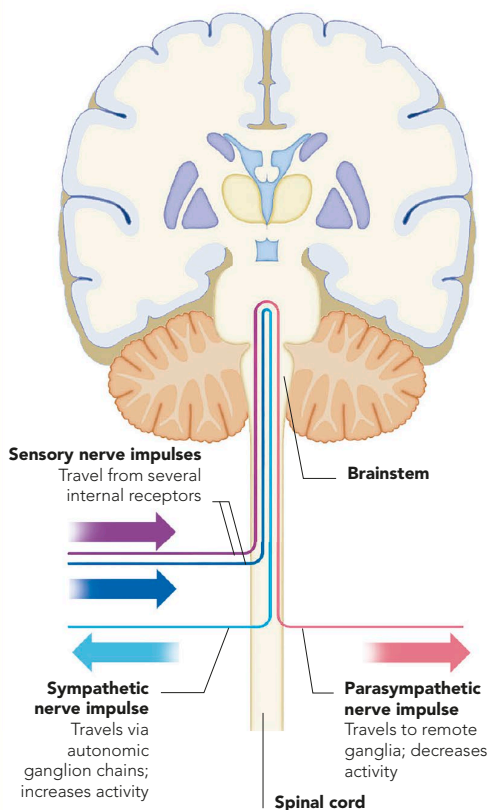


INVOLUNTARY RESPONSES

There are two main types of involuntary, or automatic, responses, which do not usually involve conscious awareness.

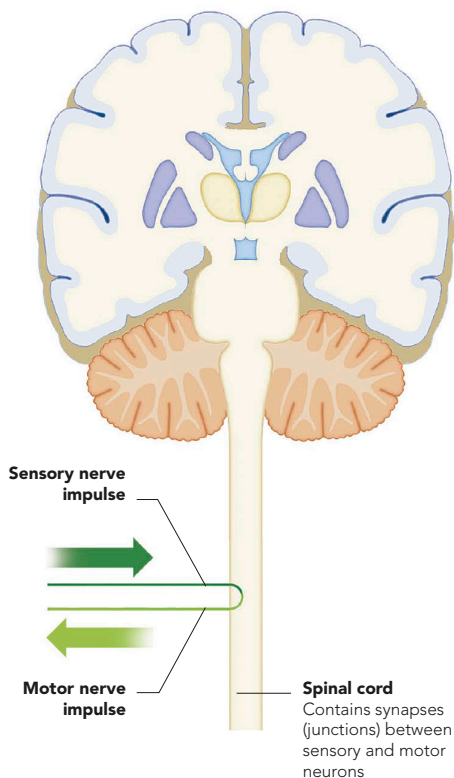
One involves reflex actions (see p.105). Reflexes mainly affect muscles normally under voluntary control. The other type of response includes autonomic motor actions. The initial nerve pathways for these responses run along spinal nerves into the spinal cord, then up ascending

nerve tracts to the lower autonomic regions of the brain, particularly parts of the limbic system and the hypothalamus. These regions analyse and process the information received, and then use the autonomic pathways to send out motor impulses, as instructions for the involuntary muscles and the glands. Response signals for the parasympathetic and sympathetic systems have separate pathways.



AUTONOMIC RESPONSES

Nerve signals pass along spinal nerves and up the spinal cord to the lower autonomic regions of the brain, which output motor impulses in response.



REFLEXES

Sensory signals arrive, and motor signals depart, wholly within the spinal cord, and without the involvement of the brain – although the brain becomes aware soon afterward.